WHAT THE SIGN SAYS

For three vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ in $\mathbb{R}^3$, the determinant of the matrix with those columns is the signed volume of the parallelepiped they span. Its sign reflects handedness.

SIGNED VOLUME

$$\det(A) = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$$

split by **the sign of** $\det(A)$

+

CASE 1

$\det(A) > 0$

Right-handed

the same orientation as the standard basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$

E.G. columns $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$: $\det = 6$

−

CASE 2

$\det(A) < 0$

Left-handed

the opposite orientation; the volume is $|\det(A)|$

E.G. the same columns with the first two swapped: $\det = -6$

0

CASE 3

$\det(A) = 0$

Flat

the three vectors are coplanar and the parallelepiped has no volume

TAKEAWAY

Swapping two columns flips the sign but not the volume: the box with $\det = 6$ becomes $\det = -6$, left-handed instead of right-handed.